

Lab05-Polymorphism

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VNU-HCM, HCMUS, FIT, SE

CS202 - Programming Systems

25A01 - 02

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A: YY: 01

H: YY: 06



Students are permitted to use AI assistance for this series of homework.

If you do not use AI: Please include Declaration A in your README.md file: "I did not use AI for this homework."

If you use AI: Please include Declaration B in your README.md file: "I did use AI for this homework." The expected workflow is that you prompt the AI, it generates a plan, you review that plan, the AI executes the action, and you review, edit, and re-prompt as needed. Ultimately, you bear full responsibility for the final result.

Design class diagrams using Mermaid format and implement the required functionality in C++ for the following assignments. For your final submission, please ensure you include the C++ source code, the raw Mermaid syntax, and the rendered image files for each class diagram.

Class diagram: https://en.wikipedia.org/wiki/Class_diagram

Use stereotypes to annotate the meaning of classes and methods.

```
<<interface>>, <<abstract>>, <<concrete>>, <<constructor>>,  
<<destructor>>, <<friend>>, <<static>>, <<virtual>>, <<pure  
virtual>>, <<override>>, <<operator>>.
```

```
Ex: +<<pure virtual>> getSalary(): double
```

Assignment 1 – Company

Company ABC asks you to build an employee management system.

The system should store the following info for each employee: employ id, full name, hire date, address.

There are 2 types of employees in the company: Office employee and worker.

- Salary of an office employee in a month = number of his/her working days * pay rate per day (300.000 VND/day)
- Salary of a worker in a month = number of items he/she produced in the month * rate pay per item (5.000VND/item)

In this week exercise, you are asked to create 3 classes: Employee, OfficeEmployee and Worker. OfficeEmployee and Worker extends/inherits from Employee.

Then you must define and implement the following methods for each Employee classes.

1. Input the information of an employee
2. Print the information of an employee to console.
3. Compute the salary in a month of an employee
4. Define at least 5 constructors for each of 3 classes.

Then, define another class, Company, **store all types of employees in a single array**. And write the following methods for class Company:

1. Input a list of employees.
2. Print out the info of all employees in the company.
3. Print out the total salary the company has to pay in a month.

4. Print out all employees whose salaries are the biggest value in the company.

Assignment 2 – Console

What is printed to the console? Give a brief explanation.

Assignment 3

```
#include <iostream>
using namespace std;
class A {
private:
    char *m_s;
public:
    A() { m_s = strdup("default"); }
    A(char *s) { m_s = s; }
    virtual void prepare() { cout << "A "; }
    void display() {
        prepare();
        cout << m_s << endl;
    }
};
class B : public A {
public:
    B(char *s) : A(s) { }
    B(const B &b) { }
    void prepare() { cout << "B "; }
};
void foo(A *obj1, A obj2) {
    obj1->display();
    obj2.display();
}
void main() {
    B obj1("text");
    A *obj2 = new B(obj1);
    foo(&obj1, *obj2);
}
```

1. What is printed to the console? Give an brief explanation.
2. Identify the memory issues in the above program and the correct them.

3. Write the input and output operators for class A and class B.

Submit a report for Questions 1 and 2, and provide the full source code for Question 3.

Assignment 4 – Prince and Princess

A beautiful princess has been imprisoned by a witch in a fortified castle with N layers of gates. To enter the castle and rescue the princess, the prince must pass through all these gates. Each gate is guarded by a gatekeeper, and there are three types of gates.

- **Business Gate:** The gatekeeper here is a merchant. To pass through, the prince must buy an amount of products from the merchant, with the cost calculated as unit price * quantity.
- **Academic gate:** The gatekeeper here is a sage. To pass through, the prince must answer the sage's question. The question has a specified intelligence level and the prince's IQ level must be equal to or greater than this level to answer correctly. Of course, answering a question does not reduce the prince's IQ.
- **Power gate:** The gatekeeper here is a warrior. To pass through, the prince must defeat the warrior. The warrior has a strength level, and the prince must have a strength level equal to or greater than the warrior's to succeed. After winning the battle, the prince's strength level will be slightly reduced.

Students are asked to apply OOP principles (encapsulation, data hiding, inheritance, polymorphism) to create a class diagram and write a C++ program that performs the following tasks:

1. Input the prince's initial parameters: money, IQ level, strength level, and the information of the castle with n gates.
2. The program will determine if the prince can rescue the princess. If he can, the program should display the prince's remaining parameters. If not, it should display a list of remaining gates.

Assignment 5 – Inheritance in Python

Write a presentation (Markdown format) with at least 20 slides to explain the following OOP concepts in Python:

- Polymorphism

Assignment 6 – Inheritance in Java

Write a presentation (Markdown format) with at least 20 slides to explain the following OOP concepts in Java:

- Polymorphism